Setup

The fundamental result we rely on is *the Bolzano–Weierstrass theorem*, which says every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

We also use the special case *the mean value theorem* for continuous functions on closed intervals.

Application

The optimisation argument splits into two halves. The first half appeals to *the Bolzano–Weierstrass theorem (p. 1)* to extract a convergent subsequence from the candidate set, then the second half uses *the mean value theorem (p. 1)* to locate the maximiser.

Concretely, the algorithm is well-defined whenever the input space is bounded — exactly the condition *the Bolzano–Weierstrass theorem (p. 1)* needs.

For the proof outline, see also the *Setup (p. 1)* section above.

A look back

Three cross-reference styles appear in this document:

- inline anchor on a `<span>` carrying the term itself (*the Bolzano–Weierstrass theorem (p. 1)*, captured at character level)
- block anchor on a heading (*Application (p. 1)*)
- the anchor's text and the page number both come from CSS, so removing or moving a section updates every reference automatically on the next render pass.